JULIAN ESTEBAN RINCÓN RODRÍGUEZ

Computer Science & Artificial Intelligence

Bogotá, Colombia • julianer2002@gmail.com • +57 311 828 3342 • github.com/Julian-Rincon

PROFESSIONAL PROFILE

9th-semester (final) Computer Science & AI student at Universidad Sergio Arboleda, specialized in data science, machine learning, and intelligent systems. Hands-on experience building data pipelines, training predictive models (XGBoost, LightGBM, neural networks), cloud infrastructure on AWS, and local AI applications. Proven ability to take projects beyond the academic setting and deliver production-grade, documented work.

EDUCATION

Universidad Sergio Arboleda — Bogotá, Colombia *2021 – Present*

B.Sc. Computer Science & Artificial Intelligence | 9th semester of 9

GPA: 4.04/5.0 (Colombian scale) · German equivalent ≈2.4 (Modified Bavarian Formula) · Upward trend: last 3 semesters avg. 4.5+/5.0

Completed coursework: Big Data & Data Engineering, Machine Learning, Simulation & Modeling, Neural Networks (Classification, Elective), Engineering Communication, Ethics & Social Responsibility.

FEATURED PROJECTS

NEXUS — Autonomous Personal AI System *2025 – Present*

Python 3.11 · 5-tier LLM routing (Cerebras/NVIDIA NIM) · GCP hybrid deploy · Custom wake word · Cloned voice (ElevenLabs) · MCP server

Independent project, closed-source by design: the public repo documents the architecture, not the code.

- Architected 5-tier LLM routing (Ollama → Groq → Cerebras 2600 tok/s → NVIDIA NIM 128K-context multimodal → Gemini fallback) with 38 auto-discovered skills spanning home automation, personal finance parsing, and passive security monitoring.
- Built an MCP server exposing system capabilities as tools for external AI agents; deployed a hybrid cloud/local architecture on a free-tier GCP VM (Tailscale-tunneled, service-account impersonation) running core services 24/7 with automated cost-guard cutoff.
- Designed safety-first automation: confirmation-gated system power control and allowlisted actions ensure irreversible commands never fire on a single ambiguous phrase; custom-trained wake word (ONNX) and cloned voice via ElevenLabs with local fallback.
- Vector memory via ChromaDB for context persistence; real home automation (IoT) and personal finance analysis active in production.

SAVI v2 — Autonomous Real Estate Valuation System *2026*

Team project (with Valeria Larea, Nicolás Garzón, and Juan Niño) for the Machine Learning course.

Python · XGBoost · K-Means · MDP · Q-Learning · Deep Q-Network · PyTorch · scikit-learn

- Architected full RL decision pipeline: MDP Value Iteration (convergence 259 iter, $\gamma=0.95$) → Q-Learning tabular (8K episodes, ε-greedy decay) → Double DQN (PyTorch, 150 epochs, replay buffer 20K, Adam lr=3e-4).
- Consensus policy across all three agents autonomously APPROVE/REVIEW/REJECTs valuations based on expected financial risk; XGBoost + cluster feature achieves $R^2=0.9609$ on 20,203 properties (2006–2024).
- Delivered IEEE-format technical paper, interactive HTML demo on GitHub Pages, and a full reproducible Python pipeline.

Complex Systems Simulators *2024*

Python · Cellular Automata · SEIR · Runge-Kutta 4 · FSM

- incendios_agentes.py: forest-fire simulator with firefighter agents (FSM) and queueing metrics.
- epidemia_dinamicos.py: SEIR+H epidemiological model with RK4/Euler numerical integration; full technical documentation delivered to the team.

Neural Networks & RL Labs *2026*

PyTorch · TensorFlow · CUDA · CIFAR-10 · OpenAI Gym

- Reinforcement learning: Q-learning (FrozenLake), REINFORCE (CartPole), and DQN with replay buffer.
- CNN labs: manual convolution, SmallCNN with dropout and weight decay trained on CIFAR-10; professional Word reports.

ShopStream — End-to-End AWS Big Data Pipeline *2026*

Python · PySpark · AWS Lambda · AWS EMR · AWS Glue · RDS PostgreSQL · Flask · Zappa

- Engineered a production-grade AWS pipeline for an e-commerce platform: 2.5M synthetic events generated → partitioned ingestion into S3 → Lambda validation (CloudWatch metrics + quarantine) → EMR/PySpark analytics (6 KPIs + anomaly detection) → Glue ETL → RDS PostgreSQL DWH → Flask API via Zappa.
- 86% test coverage with pytest; CI/CD via GitHub Actions; RDS security groups hardened for controlled access.

Project Dogma — Stochastic Social Propagation Engine *2026*

TypeScript · React · Dynamic Markov Chains · Hidden Markov Model · Nash Equilibrium · Monte Carlo

- Mathematical simulation engine for ideological propagation across urban social graphs: Dynamic Markov Chains model agent state transitions (Neutral → Believer → Fanatic), with per-agent solitude, skepticism, and charisma variables modifying the transition matrix each tick.
- Authority behavior modeled via Hidden Markov Model (Forward algorithm): 4 latent states inferred from a single observable signal (Heat entropy); doctrinal conflicts resolved via 2×2 Nash Equilibrium mixed strategy (55%/45%), with resource economy governed by the Verhulst logistic differential equation.
- Validated across 500 Monte Carlo simulations: $\mu=152$ ticks, conversion 5.42 agents/tick ($\sigma=0.39$), retention ratio 38:1.

Big Data Platform — AWS (Chinook / Sakila / Credit) *2026*

Chinook: team project (with Juan Hurtado and David Martinez) for the Big Data course.

AWS EC2 · RDS (PostgreSQL / MySQL) · Athena · S3 · AWS Glue · Terraform · React · FastAPI · Git

- Fullstack on Chinook DB: React/Vite frontend + FastAPI backend on separate EC2 instances, PostgreSQL on RDS; 17-section technical document.
- Loaded Sakila DB into RDS MySQL, resolving the log_bin_trust_function_creators restriction via a Custom Parameter Group.
- Hive-partitioned external table (credit2) on Athena + S3; resolved a case-sensitivity issue in partitions via manual ALTER TABLE.
- NYC Yellow Taxi dataset (94M rows, 12 months) on AWS EMR with PySpark; resolved schema inconsistency across Parquet files using explicit StructType and per-file normalization with unionByName.

Firma360 — Data Science Portfolio *2025 – Present*

Python · Faker · openpyxl · ETL · Power BI · Ollama

- 360° reporting prototype for corporate clients using synthetic Colombian data and an automated ETL pipeline.
- Modules: generar_datos.py, etl.py, exportar_excel.py; Power BI dashboards; long-term vision of local LLM querying via Ollama.

Custom Programming Language *2023*

Python · ANTLR4 · Tokenization · Interpretation

- Designed and implemented a Python-inspired language with a custom tokenizer, lexer, and parser, and a visitor-pattern execution engine, using ANTLR4.

Distributed Mesh Network with B.A.T.M.A.N Protocol *2023*

Linux · Networking · Distributed Systems

- Functional mesh network across four devices with dynamic routing and performance metrics.

Parallel Image Processing with CUDA *2023*

CUDA · C · Matrix operations · GPU

- Reduced-time image transformation via GPU-parallelized matrix operations.

TECHNICAL SKILLS

Languages	Python (intermediate–advanced), Bash Scripting, JavaScript (basic), SQL
ML / AI	scikit-learn, XGBoost, LightGBM, PyTorch, TensorFlow, Whisper, Ollama
Big Data & Cloud	AWS (EC2, RDS, S3, Athena, Glue, EMR), Terraform, Git / GitHub, Docker (basic)
Data & ETL	Pandas, NumPy, Matplotlib, openpyxl, Power BI, Jupyter Notebook, PySpark
Automation	Playwright, pywinauto, Telegram Bot API, Gmail / Calendar OAuth2
Other	ANTLR4, CUDA, NetworkX, ChromaDB, ZeroMQ, tkinter, pytest

ACHIEVEMENTS & RECOGNITION

International Award Winner — WarmiTics (Wisibilízalas, Universidad Pompeu Fabra) *2019*

Universidad Pompeu Fabra · Barcelona, Spain · Shakespeare School, Quito, Ecuador · Senior Category

- 1st place in an international competition organized by Universidad Pompeu Fabra, representing Shakespeare School (Ecuador) in the Senior category at age 17, against teams from Latin America and Europe.
- As Technical Product Manager and Lead Frontend/UX, architected and developed the full web application; managed cross-functional logistics; conducted structured interviews with women leaders in technology to build the site's primary content.
- Project available at: sites.google.com/view/warmitics

1st Place — USABOT Robotics Challenge

- Led hardware design and commercial strategy for an intelligent automatic irrigation system, winning 1st place against competing university teams. Designed the 3D CAD model for the physical prototype, integrated ultrasonic sensor architecture for real-time water-level detection and automated valve control, and wrote and delivered the go-to-market pitch that won over the judges.

Independent Project — NEXUS *2025 – Present*

Self-directed development outside the academic curriculum

- Autonomous personal AI system built from scratch, manages the machine, not just conversation. 38 skills, MCP server, hybrid cloud/local deploy (GCP), safety-gated automation. Personal initiative beyond the academic curriculum.

CERTIFICATIONS

Database Design — Oracle Academy — August 2024

Database Programming with SQL — Oracle Academy — December 2024

AWS Academy Graduate — Data Engineering — Amazon Web Services Academy — May 2026 — 40 hours · Credly: credly.com/go/UGp1CUEB

LANGUAGES

Spanish: Native • English: B2+ (technical and academic use: documentation, presentations, reports)

SOFT SKILLS

- Technical leadership: led CAD design and sensor architecture for USABOT (1st place), including the commercial pitch delivered to judges.
- Effective communication: interviewed women leaders in technology for WarmiTics, translating their stories into content for an international audience (1st place, Wisibilízalas).
- Teamwork: coordinated with 3-to-4-person teams on SAVI v2 and the Big Data Platform (AWS), both collaboratively built course projects.
- Analytical thinking: designed the multi-agent consensus policy (MDP + Q-Learning + DQN) behind SAVI v2 for financial risk decisions.
- Self-directed learning: built NEXUS, a personal AI system, entirely outside the academic curriculum.
- Commitment to continuous improvement: upward-trending GPA, last 3 semesters above 4.5/5.0.
- Resilience: sustained NEXUS and other independent projects alongside a full course load for the past two years.

Available for internships, freelance projects, or collaborations in AI, Data Science, and Software Development.